

Experiment Registry Report

Source: `knowledge/experiment_registry.csv` · Generated 2026-09-18 13:19 UTC · 579 runs 576 completed 3 skipped

Flat index of every registered classification run in this R&D workspace across HyperAck and 10 UCI datasets, with suite/mode/model tags and locked-split metrics.

REGISTERED RUNS	MEAN ROC-AUC	MEDIAN ROC-AUC	KNOWN LEAKAGE
579	0.913	0.936	217

Governance: 123 rows are marked both `has_known_leakage=True` and `deployment_eligible=True`. Treat `deployment_eligible` as registry metadata, not production approval. Provenance filled on only 10 rows.

1. Coverage by suite

Suite	Rows	Datasets	Contents	Mean ROC	Max ROC
external_playbook	302	10 UCI	safe/unsafe FE ladder	0.9017	1.0000
external_benchmark	140	10 UCI	7-model zoo × baseline/optimized	0.9112	1.0000
hyperack_optimized_safe	53	hyperack	tuned safe recovery / ensembles	0.9394	0.9455
hyperack_4way	44	hyperack	safexunsafe × baseline×optimized	0.9473	0.9802
hyperack_initial	15	hyperack	early exploration	0.9725	0.9793
hyperack_safe	15	hyperack	safe leakage suite	0.9389	0.9450
model_building_50_v1	10	10 UCI	campaign holdout snapshots	0.8803	1.0000

Mode mix

Mode	Rows
safe	253
unsafe	186
external	140

Runs per dataset

Dataset	Rows
hyperack	127
breast_cancer	46
heart_disease	46
adult	45
bank_marketing	45
credit_default	45
german_credit	45
mushroom	45
online_shoppers	45
spambase	45
wine_quality	45

2. Model families

Row counts

Safe-mode ROC by family

model_family	Rows
lightgbm	244
ensemble	73
extra_trees	52
xgboost	52
hist_gradient_boosting	49
logistic_regression	46
catboost	26
random_forest	25
linear_svc	5
tabular_transformer	4
tabPFN	3

Family	n	Mean ROC	Max ROC
catboost	4	0.9425	0.9435
random_forest	3	0.9412	0.9414
ensemble	47	0.9215	1.0000
xgboost	18	0.9141	1.0000
hist_gradient_boosting	16	0.9126	1.0000
tabular_transformer	2	0.9118	0.9186
lightgbm	124	0.9024	1.0000
linear_svc	3	0.8826	0.8829
extra_trees	20	0.8785	1.0000
logistic_regression	14	0.8672	0.9996

3. Dataset champions (best ROC-AUC)

Best completed score per dataset × mode. Large safe→unsafe gaps appear only where leakage features are declared.

Dataset	Safe ROC	Safe recipe	Unsafe ROC	Unsafe recipe	External ROC	External recipe
adult	0.9261	lightgbm / none / 14f	0.9261	lightgbm / none / 14f	0.9274	catboost / optimized / 14f
bank_marketing	0.8029	lightgbm / none / 15f	0.9357	xgboost / RandomizedSearchCV_roc_auc / 53f	0.9386	catboost / optimized / 16f
breast_cancer	1.0000	lightgbm / none / 69f	1.0000	lightgbm / none / 69f	0.9990	extra_trees / optimized / 30f
credit_default	0.8005	stacking_ensemble / oof_stacking / 59f	0.8005	stacking_ensemble / oof_stacking / 59f	0.7974	lightgbm / optimized / 23f
german_credit	0.8082	xgboost / RandomizedSearchCV_roc_auc / 61f	0.8082	xgboost / RandomizedSearchCV_roc_auc / 61f	0.8112	catboost / optimized / 20f
heart_disease	0.9740	hist_gradient_boosting / none / 46f	0.9740	hist_gradient_boosting / none / 46f	0.9621	hist_gradient_boosting / baseline / 13f
hyperack	0.9455	Soft3 / optimized / 25f	0.9802	hist_gradient_boosting / baseline / 34f	—	—
mushroom	1.0000	hist_gradient_boosting / none / 59f	1.0000	hist_gradient_boosting / none / 59f	1.0000	catboost / baseline / 22f
online_shoppers	0.7737	xgboost / RandomizedSearchCV_roc_auc / 59f	0.9311	lightgbm / RandomizedSearchCV_roc_auc / 61f	0.9324	lightgbm / optimized / 17f
spambase	0.9878	lightgbm / none / 57f	0.9878	lightgbm / none / 57f	0.9878	lightgbm / baseline / 57f
wine_quality	0.8968	stacking_ensemble / oof_stacking / 45f	0.8968	stacking_ensemble / oof_stacking / 45f	0.9160	extra_trees / baseline / 11f

4. Leakage gaps (unsafe – safe champions)

Dataset	Gap	Safe ROC	Unsafe ROC	Material?
adult	+0.0000	0.9261	0.9261	none
bank_marketing	+0.1328	0.8029	0.9357	material
breast_cancer	+0.0000	1.0000	1.0000	none
credit_default	+0.0000	0.8005	0.8005	none
german_credit	+0.0000	0.8082	0.8082	none
heart_disease	+0.0000	0.9740	0.9740	none
hyperack	+0.0347	0.9455	0.9802	material
mushroom	+0.0000	1.0000	1.0000	none
online_shoppers	+0.1575	0.7737	0.9311	material
spambase	+0.0000	0.9878	0.9878	none
wine_quality	+0.0000	0.8968	0.8968	none

Material drivers: **online_shoppers** PageValues; **bank_marketing** duration; **hyperack** final_customer_fare / final_biker_fare. Elsewhere safe and unsafe champions match.

5. External benchmark: baseline vs optimized

Mean ROC across 7 models per dataset. Optimization lift is small and sometimes negative.

Dataset	Baseline mean	Optimized mean	Lift
adult	0.9075	0.9107	+0.0031
bank_marketing	0.9251	0.9269	+0.0019
breast_cancer	0.9961	0.9956	-0.0005
credit_default	0.7724	0.7824	+0.0100
german_credit	0.7853	0.7841	-0.0012
heart_disease	0.9501	0.9479	-0.0022
mushroom	0.9999	1.0000	+0.0000
online_shoppers	0.9147	0.9161	+0.0014
spambase	0.9834	0.9817	-0.0018
wine_quality	0.8773	0.8670	-0.0103

6. HyperAck safe ceiling (top optimized-safe)

ROC	Experiment	Model	Optimization	Features
0.9455	softvote_etbag_lgbmwinner_xgb	Soft3	optimized	25
0.9450	stacking_top_tuned_models	StackingClassifier	optimized	25
0.9450	lgbm_prior_winner_replay	LGBMClassifier	optimized	25
0.9446	soft_avg_winner_te_optuna	FixedSoft	optimized	25
0.9446	lgbm_top7_oof_seeds_bag	TopSeedBag	optimized	25
0.9446	lgbm_best_oof_seed	LGBMClassifier	optimized	25
0.9445	lgbm_best_of_pass2_seed_bag	SeedBagLGBM	optimized	25
0.9445	softvote_top_tuned_models	SoftVoteTuned	optimized	25

Safe champion SoftVote ~0.9455 on 25 pre-decision features. Unsafe HistGB baseline with final fares reaches ~0.9802 (leakage ceiling).

7. Metric distribution (completed)

ROC RANGE 0.705–1.000	MEDIAN F1 0.848	MEDIAN FIT 1.1s	MAX FIT 874s
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Slowest fits

Fit	Dataset	Model	Optimization	Suite
873.5s	hyperack	RandomizedSearchCV(LGBMClassifier)	optimized	hyperack_optimized_safe
865.9s	hyperack	RandomizedSearchCV(LGBMClassifier)	optimized	hyperack_optimized_safe
585.5s	hyperack	RandomizedSearchCV(LGBMClassifier)	optimized	hyperack_optimized_safe
528.7s	hyperack	RandomizedSearchCV(RandomForestClassifier)	optimized	hyperack_optimized_safe
519.4s	hyperack	RandomizedSearchCV(LGBMClassifier)	optimized	hyperack_optimized_safe
299.6s	hyperack	RandomizedSearchCV(SelectKBest+LGBM)	optimized	hyperack_optimized_safe

8. Optimization labels

optimization	Rows
none	162
optimized	145
baseline	96
RandomizedSearchCV_roc_auc	40
unspecified	30
hand_tuned	20
isotonic_calibration	20
oof_stacking	20
soft_probability_blend	20
mutual_info_selection	20
extra_trees_C02	4
lightgbm_C02	1

9. Skipped runs

Dataset	Experiment	Model	Notes
hyperack	11	TabPFNClassifier	Skipped: TabPFN license/token required (set TABPFN_TOKEN).
hyperack	11	TabPFNClassifier	Skipped; no interactive TabPFN license in this environment.
hyperack	11	TabPFNClassifier	Skipped in safe suite for the same license reason as unsafe suite.

10. Schema & data quality

31 columns — identity, taxonomy, governance, shape, metrics, cost, notes - All run_id / fingerprint values unique (no duplicate registrations) - roc_auc / AP / accuracy / recall / f1 / fit_seconds: 99.5% - train_rows / test_rows: 78.1%	Source path roots - general_pipeline/results_external — 140 - external_projects/*_exp — 302 - optimized_safe_model/results — 53 - general_pipeline/results (4-way) — 44 - hyperack_exp + safe_leakage_exp — 30
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- notes: 66.1% · strategy: 14.3%
- schema_version / provenance: 1.7% (campaign only)

- campaigns/model_building_50_v1 — 10

11. Takeaways

1. Registry indexes **579** distinct runs across HyperAck + 10 UCI datasets.
2. Material leakage lift is concentrated in bank_marketing, online_shoppers, and HyperAck.
3. Safe HyperAck ceiling ~**0.9455**; unsafe leakage ceiling ~**0.980** — do not promote leakage-tagged scores.
4. External “optimized” rarely beats baseline by more than ~0.01 mean ROC; wine_quality regresses under optimization.
5. Densify schema_version / has_provenance on future ingests; fix deployment_eligible for leakage rows.